DID EPRC Workshop Talk

Working Slide Outline With Planned Figures and Tables

Draft working document

October 2026

Overview

This document drafts the main content for each slide in the DID EPRC workshop talk. It is not the final slide deck. Each entry gives the slide-facing text and then shows the table, equation, or figure that would appear on that slide.

The empirical motivation is Strickland, Chan, and Gottfried (2026), an *Educational Evaluation and Policy Analysis* paper on NYC's Gifted and Talented (G&T) program using a staggered difference-in-differences design. The workshop exercises use synthetic panel data generated for teaching, inspired by that setting; they are not the NYC administrative data used in the paper.

Part A. DID Setup and Intuition

Slide 1. Title and Roadmap

Main.tex 1: DID is popular in policy and program evaluation, and TWFE is commonly used.

Difference-in-differences (DID) is widely used in education policy and program evaluation when random assignment is not feasible and some districts are exposed to a policy while others are not. The roadmap makes clear that the talk moves from traditional DID to TWFE, event studies, staggered adoption, and then hands off before the covariates section.

Approx. time	Topic
0–5 min	Why DID is useful for education policy questions.
5–15 min	Main.tex 1(a): traditional DID and parallel trends.
15–25 min	Main.tex 1(b): TWFE, event studies, and TWFE limitations.
25–35 min	Main.tex 1(c): staggered DID example and TWFE problem.
35–45 min	Main.tex 1(c)(ii): CSA-style estimator, aggregations, inference, and alternatives.

Planned table for Slide 1: roadmap keyed to the main.tex outline.

Slide 2. Policy Example: NYC Gifted and Talented

Main.tex 1(a), setup: use one education-policy example to motivate traditional DID.

Policy example / research question. NYC's Gifted and Talented (G&T) program motivates the workshop example. Strickland, Chan, and Gottfried (2026) study the longitudinal effects of NYC's G&T program using a staggered DID design.

Synthetic-data research question. Do districts that implement a G&T-style program in year 5 experience larger gains in average achievement than districts that do not implement the program?

Important clarification. The workshop data are synthetic. They are inspired by this education policy setting, but they are district-level teaching data, not the NYC administrative data used in the paper.

DID piece	G&T example	Synthetic workshop data
Unit	Student, school, or district	District.
Time	School year or grade	Year.
Treatment	G&T participation or access	G&T-style program begins.
Treated group	Students or settings exposed to G&T	Districts treated in year 5.
Comparison group	Not-yet or never exposed students/settings	Never-treated districts.
Outcome	Achievement, attendance, or behavior	District average standardized achievement.

Planned table for Slide 2: policy example mapped to the synthetic data.

Slide 3. Traditional DID Setup

Main.tex 1(a): one example of traditional DID.

In the simplest DID design, we compare two groups across two periods. One group receives treatment after the pre-period. The other group is not treated. DID asks whether the treated group changes more after treatment than the comparison group changes over the same period. In the synthetic data, the treated group is districts that implement the G&T-style program in year 5.

$$\widehat{\delta}_{DID} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre}).$$

Concept. DID is not just treated versus comparison after treatment. It is the treated group's change minus the comparison group's change.

Group	Pre	Post	Change
Treated	Mean before year 5	Mean after year 5	Treated change.
Comparison	Mean before year 5	Mean after year 5	Comparison change.
DID			Treated change minus comparison change.

Planned table for Slide 3: two-group, two-period DID structure.

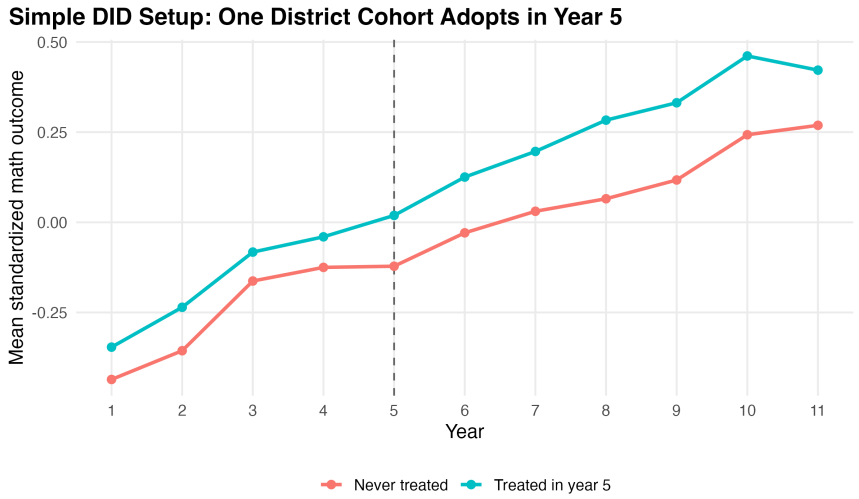
Slide 4. Parallel Trends: The Identifying Assumption

Main.tex 1(a): use a graph to show the idea of parallel trends.

Key identifying idea. DID asks what would have happened to treated units after treatment if they had not been treated. The comparison group is useful if it tells us what would have happened to the treated group in the absence of treatment. Parallel trends is therefore an assumption about untreated potential outcomes, not about the observed post-treatment outcomes.

$$E[Y_{T,post}(0) - Y_{T,pre}(0)] = E[Y_{C,post}(0) - Y_{C,pre}(0)].$$

In words: without the G&T-style program, treated districts would have experienced the same average change in achievement as comparison districts.



Planned figure for Slide 4: raw group trends from the synthetic data. Treated units adopt in year 5; comparison units are never treated.

Slide 5. Classic DID Picture: Observed vs. Counterfactual

Main.tex 1(a): use a traditional DID graph to show observed treated outcomes, the counterfactual trend, and the DID effect.

This is the classic DID teaching figure. The treated group has an observed post-treatment outcome, but the missing object is the treated group's counterfactual post-treatment outcome: where would the treated group have been without treatment?

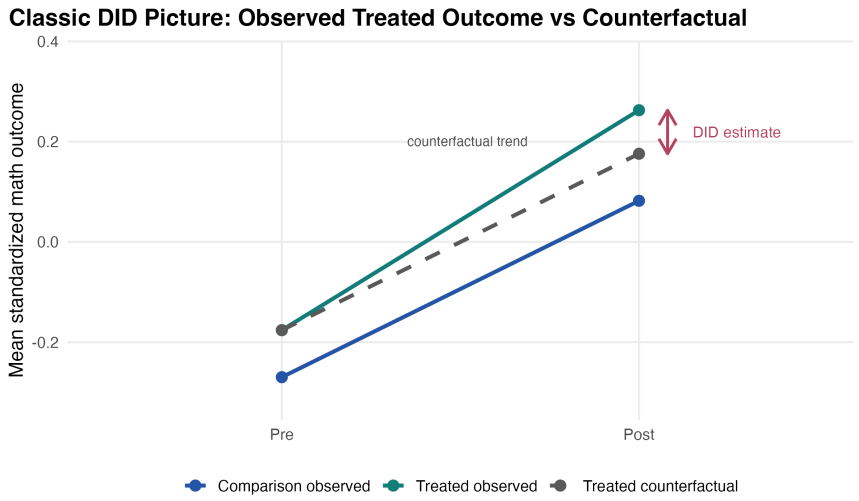
Potential outcomes. For district i in year t, $Y_{it}(1)$ is the potential outcome with the program and $Y_{it}(0)$ is the potential outcome without the program. For treated districts after year 5, we observe $Y_{it}(1)$, but we do not observe $Y_{it}(0)$. DID uses the comparison group trend to approximate that missing counterfactual.

$$\widehat{Y}_{T,post}(0) = \bar{Y}_{T,pre} + (\bar{Y}_{C,post} - \bar{Y}_{C,pre}).$$

$$\widehat{\delta}_{DID} = \bar{Y}_{T,post} - \widehat{Y}_{T,post}(0).$$

$$ATT = E[Y_i(1) - Y_i(0) \mid G_i = 1].$$

Standard DID usually targets the average treatment effect on the treated (ATT): the average effect for districts that actually implemented the program.



Planned figure for Slide 5: comparison trend, observed treated trend, dashed treated counterfactual trend, and the DID estimate.

Slide 6. DID as a Regression

Main.tex 1(a), bridge: connect the traditional DID example to the regression version before introducing TWFE.

The same two-by-two contrast can be estimated with a regression. The coefficient on the interaction between treatment group and post period is the DID estimate. Because the data are generated, we can compare the estimated effect with the known simulated effect.

$$Y_{it} = \alpha + \beta Treat_i + \gamma Post_t + \delta(Treat_i \times Post_t) + \varepsilon_{it}.$$

Term	Meaning
$Treat_i$	Treated units differ from comparison units.
$Post_t$	Outcomes differ after treatment begins.
$Treat_i \times Post_t$	Treated units change more or less after treatment.
δ	DID estimate.

Planned table for Slide 6: terms in the two-by-two DID regression.

Model	Estimated effect	Known simulated effect
Two-by-two DID regression	0.087	0.090

Planned table for Slide 6: estimated DID effect versus the known simulated effect.

Part B. TWFE and Event Studies

Slide 7. Introduce the TWFE Model

Main.tex 1(b)(i): introduce the TWFE model.

Many applied DID papers estimate a two-way fixed effects (TWFE) model. Instead of only using one pre-period and one post-period, TWFE uses the full panel and controls for unit fixed effects and time fixed effects.

$$Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \varepsilon_{it}.$$

Concept. TWFE is the fixed-effects version of the DID regression. In the simple one-treatment-cohort case, the treatment indicator is the same treated-after-treatment idea as $Treat_i \times Post_t$. The model is attractive because it uses repeated observations and absorbs stable district differences and common year shocks.

Model piece	Interpretation
α_i	Unit fixed effects: stable differences across districts.
λ_t	Year fixed effects: shocks common to all districts in a year.
D_{it}	Treatment indicator that equals 1 for treated districts after treatment begins.
δ	Estimated treatment effect in the simple setup.

Planned table for Slide 7: pieces of the TWFE model.

$$\widehat{\delta} = (\overline{Y}_{T,post} - \overline{Y}_{T,pre}) - (\overline{Y}_{C,post} - \overline{Y}_{C,pre}).$$

In a balanced panel with one common treatment start and never-treated comparison districts, the TWFE coefficient recovers the familiar DID contrast. The pre and post means can average over multiple years, not just one pre year and one post year.

Slide 8. Event Studies: Re-centering Time Around Treatment

Main.tex 1(b)(ii): introduce event-study models.

An event study replaces the single treatment indicator with indicators for time relative to treatment. This lets us look at outcomes before and after treatment. In the synthetic data, event time 0 is year 5, when treated districts implement the G&T-style program.

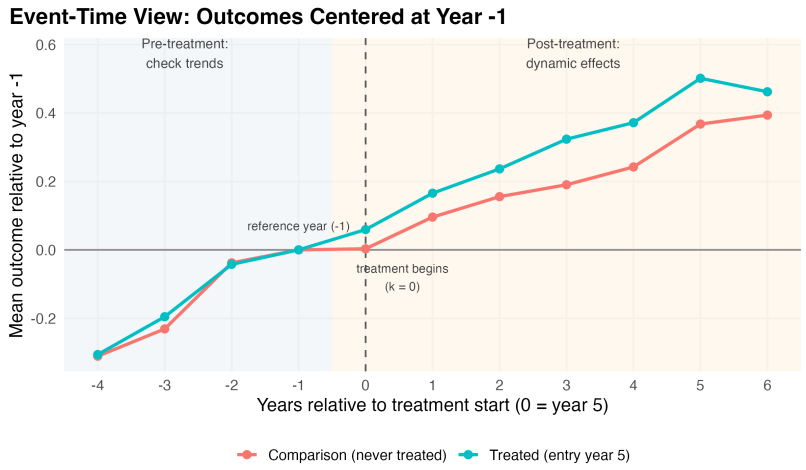
$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}[t - G_i = k] + \varepsilon_{it}.$$

Here, G_i is the year district i first receives treatment and $k = t - G_i$ is event time. The indicator $\mathbf{1}[t - G_i = k]$ equals 1 when the district-year observation is exactly k years from treatment, and 0 otherwise. Never-treated districts have all event-time indicators equal to 0.

Calendar year t	1	2	3	4	5	6	7
Treatment year G_i	5	5	5	5	5	5	5
Event time $k = t - G_i$	-4	-3	-2	-1	0	1	2

Event time $k = -1$ is omitted because it is the reference period: the year immediately before treatment. For example, for a district treated in year 5, $\mathbf{1}[t - G_i = 2] = 1$ in year 7 because $7 - 5 = 2$.

Concept. Pre-treatment coefficients show whether treated and comparison districts were already diverging before treatment. Post-treatment coefficients show whether effects appear immediately, grow, or fade after treatment.



Planned figure for Slide 8: outcomes centered at event time -1. The plot should label $k = -1$ as the reference year, $k = 0$ as treatment begins, the left side as pre-treatment, and the right side as post-treatment.

Slide 9. Limitation of a Single TWFE Coefficient

Main.tex 1(b)(iii): explain the limitation of TWFE models - one constant ATT can hide dynamic or heterogeneous effects.

A single TWFE coefficient is simple, but it compresses many possible effects into one number. This can be misleading if effects grow, fade, or appear only several years after treatment.

In the simple design, TWFE is close to the familiar DID contrast. Once treatment timing varies, TWFE can compare newly treated units to never-treated units, not-yet-treated units, and already-treated units in ways that are harder to interpret.

What TWFE gives us	**What it can hide**
One treatment coefficient	Effects that change over time.
A simple average	Differences across treatment cohorts.
A familiar regression model	Comparisons involving already-treated units.

Planned table for Slide 9: why one TWFE coefficient can be too compressed.

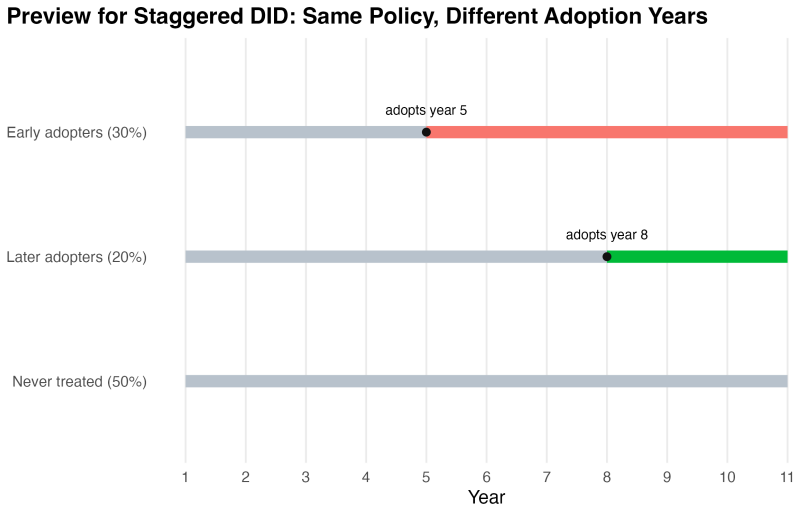
Part C. Staggered DID Preview

Slide 10. Staggered DID Example: G&T Entry at Different Times

Main.tex 1(c): one example of staggered DID, G&T, with one graph to show different treatment timing.

In many education settings, treatment does not start for everyone at the same time. Students may enter G&T at different grade points. Districts or schools may adopt a policy in different years.

In the synthetic data, one treated group adopts in year 5 and another adopts in year 8. A third group is never treated. This mirrors the core staggered-adoption problem: treatment timing is itself part of the design.



Planned figure for Slide 10: early adopters, later adopters, and never-treated units.

Slide 11. Why Staggered Timing Complicates TWFE

Main.tex 1(c)(i): discussion of the problem of TWFE - Bacon decomposition.

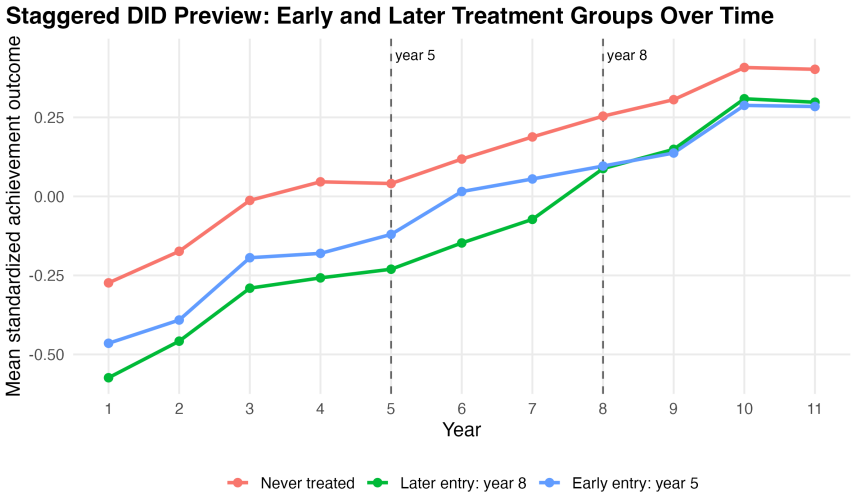
With staggered timing, TWFE no longer compares one treated group to one untreated group. Instead, the regression uses variation in treatment status across groups and years. That means the comparison group changes depending on which cohort and time period we are looking at.

In the synthetic data, there are three treatment-timing groups:

$$G_i = 5 \quad \text{early adopters}$$

$$G_i = 8 \quad \text{later adopters}$$

$$G_i = \infty \quad \text{never-treated districts.}$$



Planned figure for Slide 11: early-treated, later-treated, and never-treated group trends. The figure helps students see when each group is untreated, newly treated, or already treated.

Goodman-Bacon decomposition. With staggered adoption, the TWFE coefficient can be written as a weighted average of many smaller two-by-two DID comparisons.

$$\widehat{\beta}_{TWFE} = \sum_j w_j \widehat{\beta}_j^{2\times 2}.$$

The key question is: what comparisons are inside that weighted average? Some comparisons are intuitive; others are problematic.

Comparison	Years used	Interpretation
$G = 5$ vs. never	Years 5–11	Early adopters compared with districts that are never treated. This is the most intuitive DID comparison.
$G = 8$ vs. never	Years 8–11	Later adopters compared with districts that are never treated. This is also an intuitive DID comparison.
$G = 5$ vs. $G = 8$	Years 5–7	Early adopters compared with later adopters before the later adopters are treated. The later group is not-yet-treated, so this can be a useful comparison.
$G = 8$ vs. $G = 5$	Years 8–11	Later adopters compared with early adopters after the early adopters are already treated. This is the problematic comparison.

Planned table for Slide 11: examples of the comparisons inside a TWFE estimate with staggered treatment timing.

The problematic comparison is sometimes called a **forbidden comparison**: TWFE may compare newly treated districts to already-treated districts. But already-treated districts are not untreated controls. If treatment effects evolve over time, those districts may already have accumulated treatment effects.

This is why the single TWFE coefficient can become hard to interpret. It is not simply "the average effect of the G&T-style program." It is a weighted average of several DID comparisons, and the weights can be unequal, hard to interpret, or even negative.

Takeaway. In staggered DID, the problem is not only that there are multiple treatment timings. The problem is that TWFE chooses the comparisons and weights for us. Modern staggered DID estimators try to estimate cohort-time effects directly and then aggregate them using clearer choices.

Slide 12. Staggered DID Estimator: CSA Logic

Main.tex 1(c)(ii)(A): CSA - group units into 2-by-2 designs, estimate ATT(g,t), then take a weighted average.

TWFE mixes many comparisons together. Callaway and Sant'Anna-style staggered DID starts by estimating the pieces first: treatment effects separately by treatment cohort and calendar year.

$$ATT(g,t) = E[Y_t(g) - Y_t(0) \mid G_i = g], \qquad t \geq g.$$

Here, $G_i = g$ means district i first receives treatment in year g, and t is the calendar year where the effect is measured. For $ATT(g,t)$ to be an actual post-treatment effect, we need $t \geq g$.

Cohort	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Y11
$G = 5$	pre	pre	pre	pre	$ATT(5,5)$	$ATT(5,6)$	$ATT(5,7)$	$ATT(5,8)$	$ATT(5,9)$	$ATT(5,10)$	$ATT(5,11)$
$G = 8$	pre	pre	pre	pre	pre	pre	pre	$ATT(8,8)$	$ATT(8,9)$	$ATT(8,10)$	$ATT(8,11)$

Planned table for Slide 12: cohort-by-year cells. Post-treatment cells are group-time effects; pre-treatment cells are not treatment effects.

Examples in our data. $ATT(5,5)$ is the effect for year-5 adopters in year 5. $ATT(5,8)$ is the effect for year-5 adopters in year 8. $ATT(8,8)$ is the effect for year-8 adopters in year 8. $ATT(8,10)$ is the effect for year-8 adopters in year 10.

$ATT(5,1)$ is not a treatment effect because year 1 is before treatment for the $G = 5$ cohort. Pre-treatment cells are useful for placebo or pre-trend checks, but they are not post-treatment ATT estimates.

Slide 13. CSA Aggregations: Group, Simple, and Event-Time ATT

Main.tex 1(c)(ii)(A)(i): group, simple, and aggregate ATT summaries.

After estimating the group-time pieces $ATT(g,t)$, the next question is how to summarize them. The aggregation should follow the research question rather than letting one regression coefficient choose the summary for us.

$$\widehat{ATT} = \sum_{g,t} w_{gt} \widehat{ATT}(g,t).$$

In words: the overall summary is a weighted average of selected group-time treatment effects. Different summaries answer different questions.

Summary	Formula idea	Example from our data	Question answered
Simple ATT	Average all post-treatment $ATT(g,t)$ values.	$ATT(5,5)$ through $ATT(5,11)$ and $ATT(8,8)$ through $ATT(8,11)$.	Average effect across treated districts and post-treatment years.
Group ATT	Average within treatment cohort.	$ATT(5)$ summarizes year-5 adopters; $ATT(8)$ summarizes year-8 adopters.	Do effects differ for early vs. late adopters?
Dynamic ATT	Average by event time $e = t - g$.	$e = 0$: $ATT(5,5)$ and $ATT(8,8)$. $e = 1$: $ATT(5,6)$ and $ATT(8,9)$.	Do effects grow, fade, or appear later?
Calendar ATT	Average within calendar year.	Year 8 can include $ATT(5,8)$ and $ATT(8,8)$.	What is the average effect in a specific year?

Planned table for Slide 13: ways to aggregate group-time effects after estimating $ATT(g,t)$.

Slide 14. Inference for Staggered DID

Main.tex 1(c)(ii)(A)(ii): inference.

Inference asks how uncertain the group-time effects and the aggregated summaries are. Once we choose the estimand, the standard errors and confidence intervals should match that estimand.

$$SE\Big(\widehat{ATT}(g,t)\Big) \qquad \text{and} \qquad SE\left(\sum_{g,t} w_{gt}\widehat{ATT}(g,t)\right).$$

Choice	Workshop explanation
Analytical / influence-function SEs	Fast large-sample standard errors for $ATT(g,t)$ and aggregated summaries.
Multiplier bootstrap SEs	A resampling approach commonly used for CSA-style estimators and confidence bands.
Pointwise confidence intervals	One interval for one estimate, such as $ATT(5,8)$.
Uniform confidence bands	Joint bands for many estimates at once, especially dynamic/event-study estimates.
Aggregation	Uncertainty from each $ATT(g,t)$ carries into simple, group, dynamic, and calendar summaries.

Planned table for Slide 14: inference choices to explain conceptually before showing software options in the lab.

Data structure	Clustering choice
District-year data	Cluster by district because the same districts are observed repeatedly over time.
School-year data, schools nested in districts	If treatment timing varies by district, cluster by district. Schools in the same district share treatment timing and district-level shocks.
Rule of thumb	Cluster at the treatment assignment level or higher.

Planned table for Slide 14: clustering examples for the main district-level data and the school-within-district extension.

Slide 15. Brief Alternatives and Reading Group Handoff

Main.tex 1(c)(ii)(B): briefly introduce alternative methods; 1(c)(ii)(B)(i): briefly introduce Wooldridge pooled OLS and ETWFE.

This workshop will not fully cover every modern staggered DID estimator. The goal is to show why we need alternatives to naive TWFE and where those alternatives fit. The reading group can return to the details.

Method	What it helps clarify
Goodman-Bacon decomposition	What comparisons TWFE is using.
Callaway and Sant'Anna	Group-time treatment effects and aggregation.
Sun and Abraham	Event-study estimates with staggered timing.
Wooldridge pooled OLS / ETWFE	Regression-based alternatives for staggered designs.

Planned table for Slide 15: methods to flag before Wei's next section.

Main.tex point intentionally skipped for now: Covariates - Wei; three-level clustering or multisite design - Wei; PT/CPT testing and possible solutions - Wei; further topics and lab session.

The introductory section closes by pointing participants to the workshop materials: the synthetic data, starter code, outline/slides, and readings. The next section can pick up with covariates and conditional parallel trends.

https://katjanestrickland.github.io/did-eprc-workshop/

References To Include

Strickland, K., Chan, W., and Gottfried, M. A. (2026). Longitudinal effects of NYC's Gifted and Talented program: Evidence from a staggered difference-in-differences approach. *Educational Evaluation and Policy Analysis*. https://doi.org/10.3102/01623737261438146

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254–277.

Callaway, B., and Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230.

Sun, L., and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), 175–199.

Wooldridge, J. M. (2021). Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators.